

δ -pCluster

- ❑ Enumerating all bi-clusters (δ -pClusters) [H. Wang, et al., Clustering by pattern similarity in large data sets. SIGMOD'02]
- ❑ Since a submatrix $I \times J$ is a bi-cluster with (perfect) coherent values iff $e_{i_1j_1} - e_{i_2j_1} = e_{i_1j_2} - e_{i_2j_2}$.
- ❑ For any 2×2 submatrix of $I \times J$, define p-score

$$p - score \begin{pmatrix} e_{i_1j_1} & e_{i_1j_2} \\ e_{i_2j_1} & e_{i_2j_2} \end{pmatrix} = |(e_{i_1j_1} - e_{i_2j_1}) - (e_{i_1j_2} - e_{i_2j_2})|$$

δ -pCluster

- ❑ A submatrix $I \times J$ is a δ -pCluster (pattern-based cluster) if the p-score of every 2×2 submatrix of $I \times J$ is at most δ , where $\delta \geq 0$ is a threshold specifying a user's tolerance of noise against a perfect bi-cluster
- ❑ The p-score controls the noise on every element in a bi-cluster, while the mean squared residue captures the average noise
- ❑ Monotonicity: If $I \times J$ is a δ -pClusters, every $X \times Y$ ($X, Y \geq 2$) submatrix of $I \times J$ is also a δ -pClusters.
- ❑ A δ -pCluster is maximal if no more row or column can be added into the cluster and retain δ -pCluster: We only need to compute all maximal δ -pClusters.